Text-to-SQL Project ## 🎯 Objective Build a Text-to-SQL application using the Gemini API. The application should allow the user to enter a question in natural language, convert it into a SQL query using Gemini, execute the query on a database, and display the result. --- ## 🔄 Project Flow ```text User Question ↓ Gemini API ↓ SQL Query ↓ Database ↓ pd.read_sql() ↓ Output ``` --- ## 🛠️ Requirements You should use: * Python * Gemini API * SQL Database * Pandas * `pd.read_sql()` --- ## 💡 Example The user enters: ```text Show me the top 5 customers by total sales. ``` Gemini should generate a SQL query such as: ```sql SELECT TOP 5 customer_id, SUM(total_amount) AS total_sales FROM orders GROUP BY customer_id ORDER BY total_sales DESC; ``` Then execute the generated query using: ```python df = pd.read_sql(query, connection) ``` Finally, display the result. --- ## 📋 Requirements ### 1. Database Use Company_SD ``` ### 2. Gemini API Use the Gemini API to convert the user's question into SQL. The prompt should include the database schema so Gemini knows the available tables and columns. Example: ```text Database Schema: customers: - SSN - fname - address Convert the following question to SQL: "Show the top 5 customers by total sales." ``` ### 3. Execute SQL Execute the generated SQL using Pandas: ```python df = pd.read_sql(query, connection) ``` ### 4. Display the Result Display: 1. User question 2. Generated SQL query 3. Query result Example: ```text Question: Show the top 5 customers by total sales. SQL: SELECT TOP 5 ... Result: customer_id | total_sales ------------|------------ 101 | 25000 205 | 21000 ... ``` --- ## ⭐ Bonus — Flask Web App Create a simple web application using: * Flask * HTML * CSS The page should contain: ```text ---------- Text to SQL ---------- Ask your question: [Show top 5 customers by sales] [Generate] Generated SQL: SELECT TOP 5 ... Result: | customer_id | total_sales | |-------------|-------------| | 101 | 25000 | | 205 | 21000 | ---------- ``` --- ## 🎯 Expected Result The final application should be able to: ```text Natural Language ↓ Gemini ↓ SQL Query ↓ pd.read_sql() ↓ DataFrame ↓ Display Result ```